

Real-time driver fatigue detection system with deep learning on a low-cost embedded system

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Highlights

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We proposed two CNN models to detect driver fatigue.

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We handled the four classes for driver's status.

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The system was tested in real time on the Nvidia Jetson Nano embedded device.

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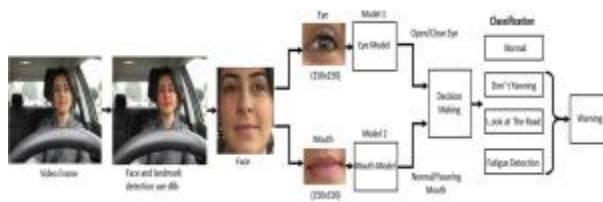
The accuracy metric for the fatigue detection for the YawDD dataset is 96%.

Abstract

Road [traffic accidents](#) result in significant life and property losses, which are caused by various factors including driver fatigue and drowsiness. Therefore, real-time monitoring of the driver's state inside a vehicle and accurate detection of fatigue is essential to reduce the number of accidents. However, achieving high accuracy with low-cost embedded devices has been a challenge. This study proposes a novel approach that uses [deep learning](#) to accurately detect driver fatigue in real-time on the Nvidia Jetson Nano embedded device. The proposed system utilizes [deep learning](#) architecture, specifically [Convolutional Neural Networks](#) (CNNs), to classify four different situations by analyzing the eye and mouth areas of the driver. In addition, the dlib library is employed to precisely locate the driver's eye and [mouth regions](#). The system is trained and tested on the YawDD dataset and achieves an accuracy of 93.6% and 94.5% for the eye and mouth models, respectively. The system operates at an average speed of 6 fps on the Nvidia Jetson Nano embedded device. The proposed system contributes to the field of driver fatigue detection by addressing the challenges of achieving high accuracy in real-time on a low-cost embedded device. This system aims to minimize the number

of accidents and protect human life during transportation by detecting driver fatigue and issuing an alert. The classification results demonstrate the success of the proposed system, which accurately classifies four different states of the driver and detects driver fatigue states with high accuracy. Overall, this study presents a significant contribution to the field of driver fatigue detection by proposing a real-time, low-cost, and accurate system that can be installed in vehicles to ensure safe transportation and prevent accidents.

Graphical abstract



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Introduction

Traffic accidents are a big social problem and cause material and moral losses. 2018 traffic accidents occurred in Turkey Referring to the distribution and characteristics; 84.81% of the accidents are material damage, 14.66% are injured and 0.53% are fatal traffic accidents [1]. In addition, TSI (Turkey Statistical Institute) by the number of vehicles in traffic according to published reports has been increasing in [2]. This increase causes an increase in traffic accidents. In addition to the increasing number of vehicles, the main factors that cause traffic accidents are the mistakes of the drivers, the carelessness of the pedestrians, and unsafe roads. Accordingly, according to the Highway Traffic Accident Statistics, it was stated that a total of 1.229.364 traffic accidents occurred in Turkey in 2018. When the flaw rates of 217.898 traffic accidents with fatal and injuries are examined, it is stated that 89.5% of the flaws are caused by the drivers [3]. In driver defects, the rate of accidents due to sleepiness while driving is 1%–16% in Europe, while this rate reaches up to 70% in Turkey [4]. According to the data of the US National Highway Traffic Safety Agency (NHTSA), there are approximately 100.000 accidents (1.5% of all accidents) and 1500 deaths (4% of all fatal accidents) each year due to fatigue and drowsiness. In the graphical image in Fig. 1, it is seen that the highest defect rate in accidents belongs to the driver with an average of 90%.

Traffic accidents can have a significant impact on society, and measures need to be taken to prevent them. Extensive research and analysis have been conducted to determine the factors that contribute to accidents, and it has been found that a high proportion of accidents are caused by driver-related factors. Driving under the influence

of alcohol, fatigue, or sleep deprivation has been identified as a significant contributor to the incidence of accidents.

It is well-known that alcohol impairs driving performance, but the dangers of driving while fatigued or sleep-deprived are often overlooked. These factors are equally dangerous and can lead to serious accidents. The negative impact of insomnia on driving performance has been demonstrated in several studies, and it has been found to be as detrimental as driving under the influence of alcohol or drugs [5].

Driver fatigue detection is a critical problem in the transportation industry. Driver fatigue and drowsiness can lead to a significant number of accidents, which can result in severe injuries and loss of life. The problem involves accurately detecting the driver's state in real-time to issue timely warnings and prevent accidents. Detecting driver fatigue is challenging due to the varying symptoms, and it is essential to accurately monitor both the face and eye regions of the driver to detect early signs of fatigue, such as drooping eyelids, yawning, and changes in facial expressions.

To prevent driver-related accidents, numerous studies have been conducted in the literature. These studies have focused on developing interventions to reduce the incidence of accidents caused by driver-related factors. Some of the interventions include the use of technology such as driver fatigue detection systems, education and training programs, and legislative measures. In recent years, deep learning methods have been increasingly employed to develop more advanced safety systems [6]. For example, deep learning algorithms have been used to improve the accuracy of speech recognition systems, which can reduce driver distraction and improve hands-free communication [7]. In the field of automotive cybersecurity, deep learning can be used to detect and prevent cyber-attacks on connected vehicles [8]. Additionally, deep learning has been applied to improve the accuracy of object detection systems, which can assist with a range of tasks such as parking assistance and obstacle detection [9]. Deep learning models have also been used in the development of real-time image enhancement for automatic automobile accident detection through CCTV [10]. Outside of the automotive industry, deep learning models have been used in a variety of applications, such as medical imaging [11], fake news detection [12], and gesture recognition [13]. In conclusion, it is crucial to explore the potential of advanced technologies such as deep learning to prevent accidents caused by driver-related factors and to consider its potential applications in other areas such as healthcare, media, and entertainment. The literature provides a range of interventions, both technological and non-technological, that can be employed to reduce the incidence of these accidents.

Girit proposed a system to analyze the video portions recorded by the camera to determine if the driver is awake or sleepy [14]. Suryaprasad and his/her friends worked on video data to detect real-time driver sleep status in their studies [15]. Parmar et al.

proposed a system that tracks the eyes and head actions to detect the drowsiness of the driver [16]. Golgiyaz et al. in their study, presented a system that decides whether to warn the driver by looking at the driver's perspective and blink status [17]. In another study, Dwivedi et al. proposed a simple CNN-based drowsy driver detection system with 78% accuracy using customized data sets [18]. In the study done by Park et al. a deep CNN of AlexNet, VGG-FaceNet and FlowImageNet networks were handled to detect driver drowsiness on RGB video [19]. When evaluating the detection system, it is stated that it works with an accuracy of 73% for NTHU-drowsy driver detection dataset. In another study, Reddy et al. provided a faster and more precise real-time drowsiness detection system with 89.5% accuracy by converting the basic model to produce a lighter model that can be placed in an embedded system [20].

Yan et al. developed a real-time system based on PERCLOS and gray scale image processing for fatigue detection [21]. Approximate position of driver's face and eye positions on gray scale images were used in the study. Deng and Wu proposed a driver fatigue system called DriCare that works with 92% accuracy in their study [22]. In the study, similar to other studies, the face and eye areas were handled and discussed. Mehta and his/her colleagues developed an Android-based real-time driver fatigue system that works with an accuracy of 84% using image processing techniques, machine learning, and the important points of the face, Eye Aspect Ratio (EAR) and Eye Closing Ratio (ECR) [23].

Galarza et al. developed a method with an accuracy rate of 93.37% [24]. In this method, the behavior of the driver's eyes, head movements and yawning situations are taken into account to detect drowsiness. In the study of Ravi et al. he proposes an automatic real-time driver fatigue detection system based on image processing technique as well as a cloud-based management platform running on Raspberry PI [25]. It was also stated that location details were learned using the GPS receiver module. Liu et al. proposed a driver fatigue detection algorithm using two stream network models with multiple facial features [26]. Multitasking cascading convolutional neural networks (MTCNNs) were used for the positioning of the mouth and eye areas. It was stated that an accuracy of 97.06% was obtained in the National Tsing Hua University Driver's Drowsiness Detection (NTHU-DDD) dataset. Savas and Becerikli proposed a multitasking Convolution Neural Network (ConNN) model for the detection of driver drowsiness in their study [27]. It was said that driver fatigue was determined by calculating the eye closing time/Percentage eye Closure (PERCLOS) and the Frequency Of Mouth (FOM). In addition, it was stated that the proposed model worked with 98.81% accuracy in YawDD and NthuDD data sets.

The present study makes a significant contribution to the field of driver fatigue detection by proposing a solution that effectively addresses the problem of detecting driver fatigue with high accuracy, including the detection of yawning, using a low-cost

embedded system. The system monitors both the face and eye regions of the driver in real time and aims to enhance transportation safety by reducing the incidence of traffic accidents and protecting human life during transportation. The proposed solution offers instant monitoring of the driver's condition and provides timely warnings to prevent accidents resulting from driver fatigue or drowsiness. This research is unique in its approach and offers valuable insights into developing low-cost, high-accuracy solutions for driver fatigue detection.

The proposed model offers several advantages and contributions to the field of driver fatigue detection. The system uses a deep learning architecture and library to accurately monitor the driver's eye and mouth regions, which can provide a more reliable measure of fatigue. The use of two separate CNN models for the mouth and eye areas also allows for more accurate detection of specific types of fatigue, such as yawning with open eyes or closed eyes. The system has been successfully implemented in real-time on an embedded device, which is a practical and convenient solution for monitoring fatigue in vehicles.

The main contribution of the proposed model is its ability to accurately detect driver fatigue in real time, which is critical for ensuring the safety of drivers and passengers. The use of deep learning techniques and the dlib library in this system demonstrates the potential of machine learning and computer vision in addressing safety concerns in the transportation industry. The proposed model also contributes to the field by presenting an approach that is more successful in detecting fatigue and yawning of drivers compared to other existing methods.

The remainder of this article is organized as follows: Section 2 provides information about the system architecture developed in this study. Section 3 presents discussion and real-time test results of the proposed system. Last section includes the conclusions and the future works regarding this paper.